

10 Chapters · 400+ Questions · 3 Quiz Levels · Free Access · No Account Required

COURSE OVERVIEW — CHAPTER BY CHAPTER

1. **Introduction** — Before IEC 61131-3, every PLC manufacturer used a proprietary language. Engineers retrained from scratch when they changed vendors. This chapter explains why the standard exists, what it unified, and which of the five languages to reach for first.
2. **Data Types** — BOOL, INT, REAL, TIME, STRING, and user-defined structured types. The type system is what separates safe PLC code from implicit conversions that silently corrupt process values at runtime.
3. **Structured Text (ST)** — Pascal-style text language: IF/THEN, FOR, WHILE, CASE, function calls. The most expressive of the five languages and the most transferable skill for engineers coming from software backgrounds.
4. **Ladder Diagram (LD)** — Relay-logic graphical representation: contacts, coils, and rungs. Electricians can read it without training. Every PLC vendor supports it. Understanding LD is non-optional for anyone working with existing automation code.
5. **Function Block Diagram (FBD)** — Signal-flow graphical language where data passes between interconnected function blocks. The natural fit for control engineers working with PID loops, signal conditioning, and process instrumentation.
6. **Sequential Function Chart (SFC)** — State machine language for batch and sequential processes. Steps, transitions, and divergence structures replace thousands of lines of LD timer-and-flag logic with a readable, maintainable flowchart.
7. **Program Organization Units** — Functions, Function Blocks, Programs, and Configurations are the module system that enables reusable automation libraries. Engineers learn how to build code that works across vendors without modification.
8. **RTAC Integration** — Applying IEC 61131-3 logic in the SEL RTAC; how the standard crosses from PLC programming into substation protection and automation. The bridge between this course and the RTAC guide.
9. **Troubleshooting** — Type mismatches, scan cycle overruns, unintended coil latching, and the runtime faults that crash programs without a clear error message. Field debugging procedures for real PLC hardware.
10. **Practice Lab** — Four exercises: a PID temperature controller in Structured Text, a motor starter sequence in Ladder, a batch filling process in SFC, and a reusable Function Block library for analog scaling.

Certification Relevance

- **PLCopen Certified** — PLCopen certification exams test IEC 61131-3 language knowledge, POU's, and programming best practices directly.
- **ISA CCST** — PLC programming, ladder logic, and control system configuration are core tested topics at all levels.
- **Vendor Credentials** — Siemens TIA Portal, Rockwell Studio 5000, and Beckhoff TwinCAT all use IEC 61131-3 as their programming foundation.

Next Steps

1. Complete IEC 61131-3 before starting the RTAC guide — logic concepts transfer directly to ACSELERATOR Architect.
2. Download CODESYS (free IDE, IEC 61131-3 compliant) and run the practice lab exercises in a soft PLC.
3. Review the PLCopen Function Block libraries for motion, communications, and safety.
4. Pursue PLCopen certification or a vendor credential (TIA Portal, Studio 5000) as the first PLC-specific credential.